NASA SPACE APPS CHALLENGE 2026

MITOCHONDRIA

Astronaut Health Intelligence Platform

A functional, integrated, open-source research platform that keeps the crew health loop working when Earth is far away — offline-first, explainable and human-supervised.

OPEN SOURCE | OFFLINE FIRST | HUMAN OVERSIGHT

https://mitochondria.edocinfo.com

Research prototype · synthetic data · not NASA-endorsed · not clinically validated · not flight-qualified

Final Competition Edition · October 2026 · Background: illustrative artwork from the project site



THE ASTRONAUT HEALTH MONITORING CHALLENGE

Spaceflight stresses the body on five interacting fronts

Space radiation

Dose and dose rate kept as separate quantities

Isolation & confinement

Fatigue, sleepiness, stress and workload as private context

Distance from Earth

Delayed or absent ground support

Altered gravity

Bone, muscle and cardiovascular adaptation

Closed environment

CO_2, O_2, pressure, temperature, noise, light

The challenge: gather health indicators so crews can evaluate and act — far from Earth.

Source: NASA Human Research Program — hazards of human spaceflight; NASA Space Apps Challenge 2026 challenge page

Distance turns health monitoring into an autonomy problem

Mission constraints → design

- Distance → local monitoring and protocol access
- Intermittent links → store-and-forward with idempotency
- Many devices → adapter contract + canonical model
- Artifacts → quality states, no silent imputation
- Adaptation → personal, phase-aware baselines
- High stakes → explainability, audit, human authority

Delayed / absent ground link

Real-time ground medicine cannot be assumed

≈ 98 ms median, 0 backend calls

Local action → visible update (validation ledger, self-check path)

Encrypt first, sync later

AES-GCM local store; authorised synchronisation

Source: Product thesis; data/validation (latency criterion)

Today’s health evidence is fragmented — and easy to over-interpret

Source	Typical problem	MITOCHONDRIA response
Wearables / phone sensors	Vendor algorithms, artifacts, missing samples	Quality levels; freshness; transport ≠ sample ≠ stored
Laboratory reports	Unstructured PDFs; units; reference ranges	Exact name + unit matching; ranges never imported
Questionnaires	Treated as diagnoses	Labelled self-report; not clinical instruments
Habitat telemetry	Implied causality	Shared timeline, explicitly non-causal
Research datasets	Mistaken for crew data	Research-only labels; no crew binding

Source: docs/CANONICAL_HEALTH_DATA.md; src/domain/biomarkers

OUR SOLUTION

One loop: Monitor → Personalize → Correlate → Detect → Explain → Act → Learn

01
Monitor
Governed adapters

02
Personalize
Phase-aware baselines

03
Correlate
One timeline, non-causal

04
Detect
Reviewed patterns only

05
Explain
Evidence, gaps, uncertainty

06
Act
Protocol categories

07
Learn
Provenance + audit

OPERATIONAL: capture, store, sync, review

RESEARCH-ONLY: interpretation

FUTURE: persisted baseline profiles

Source: src/components/app-shell.tsx (health loop); docs/HEALTH_INTELLIGENCE_V1.md

PLATFORM VALUE PROPOSITION

What the platform gives each stakeholder

Crew Self

- Local, encrypted, offline capture
- Own evidence; self-assessment
- Synthetic lab-report review

Mission Operator

- Missions, crew, invites
- Settings + redacted audit log
- Synthetic demo launchpad

Ground Medical

- Mission snapshot
- Independent synthetic review queue
- Research-only decisions

Researcher

- Aggregate-only export
- Suppressed below 3 crew / 10 events

Integrators

- Adapter SDK, canonical model
- Apache-2.0, self-host, CI

Source: Master Manual §09.5 (role-based access); research privacy workspace

Science leads — and every scientific claim carries its review state

Source	Size	Status
Biomarker Matrix v1.1 (catalogue)	70 metrics	OPERATIONAL catalogue — not clinical validation
Bone Health Handbook v7 (GR → EN)	63 variables	RESEARCH-ONLY · pending review
Cardiovascular Handbook (GR → EN)	96 + 8 examples	RESEARCH-ONLY · pending review
Immune + behavioural drafts	35 + 36	PENDING independent review
Independent science gates	8	PENDING verification

Boundaries

- No diagnosis or risk score
- No causation from association
- No clinical thresholds
- Baseline deviation ≠ diagnosis
- Questionnaires ≠ clinical instruments
- Synthetic ≠ astronaut evidence

Source: src/domain/catalog/reviewed-mvp-v1.1.ts; data/scientific/; data/validation (independent review gates)*

THE FOUR HEALTH INTELLIGENCE DOMAINS

Four health intelligence domains — each with its own review state

Bone

- Entry, CSV, document extraction
- Local evidence summary
- Handbook v7: 63 variables (EN/GR)

Cardiovascular

- Mission-scoped HR/HRV/BP context
- Handbook: 96 measurements + 8 examples
- No arrhythmia inference from PPG

Immune & microbiome

- Evidence summary + missing signals
- 35 proposals — pending review
- Microbiome–immune: research only

Behavioral, sleep & cognitive

- Readiness check-in context
- 36 proposals — pending review
- No mental-health diagnosis

Source: src/components/health-domain-workspace.tsx, bone-health-entry.tsx, scientific-handbook-reference.tsx

Biomarkers are matched exactly — relationships are shown, never inferred

Relationship	Evidence	Platform behaviour
Sleep ↔ sleepiness ↔ vigilance	supported	Research display; DATA QUALITY prompts
Workload ↔ sleep ↔ HR/HRV	supported / limited	No stress label
Bone unloading ↔ urine calcium	supported	No renal-stone score
Bone ↔ vascular (CTX, PWV…)	exploratory	Display only
Microbiome ↔ immune	research only	Never operational
CO_2 ↔ cognition	conflicting	No causal rule

Exact matching

- 159 registry entries
- 96 metric/unit pairs accepted
- 63 bone: advisory hints
- Ranges never imported
- No fuzzy or filename routing

Source: data/scientific (biomarker source registry); src/domain/health-intelligence/patterns.ts

SCIENTIFIC PROVENANCE

Every claim traces back to a file, a hash or a ledger

Scientific sources

- Greek handbooks byte-identical (SHA-256 tests)
- English layers joined by ID
- Translation ambiguities flagged for review

Evidence records

- Source SHA-256 + excerpt
- Submitter / reviewer snapshots
- Decision receipt + one event
- Audit rows without values

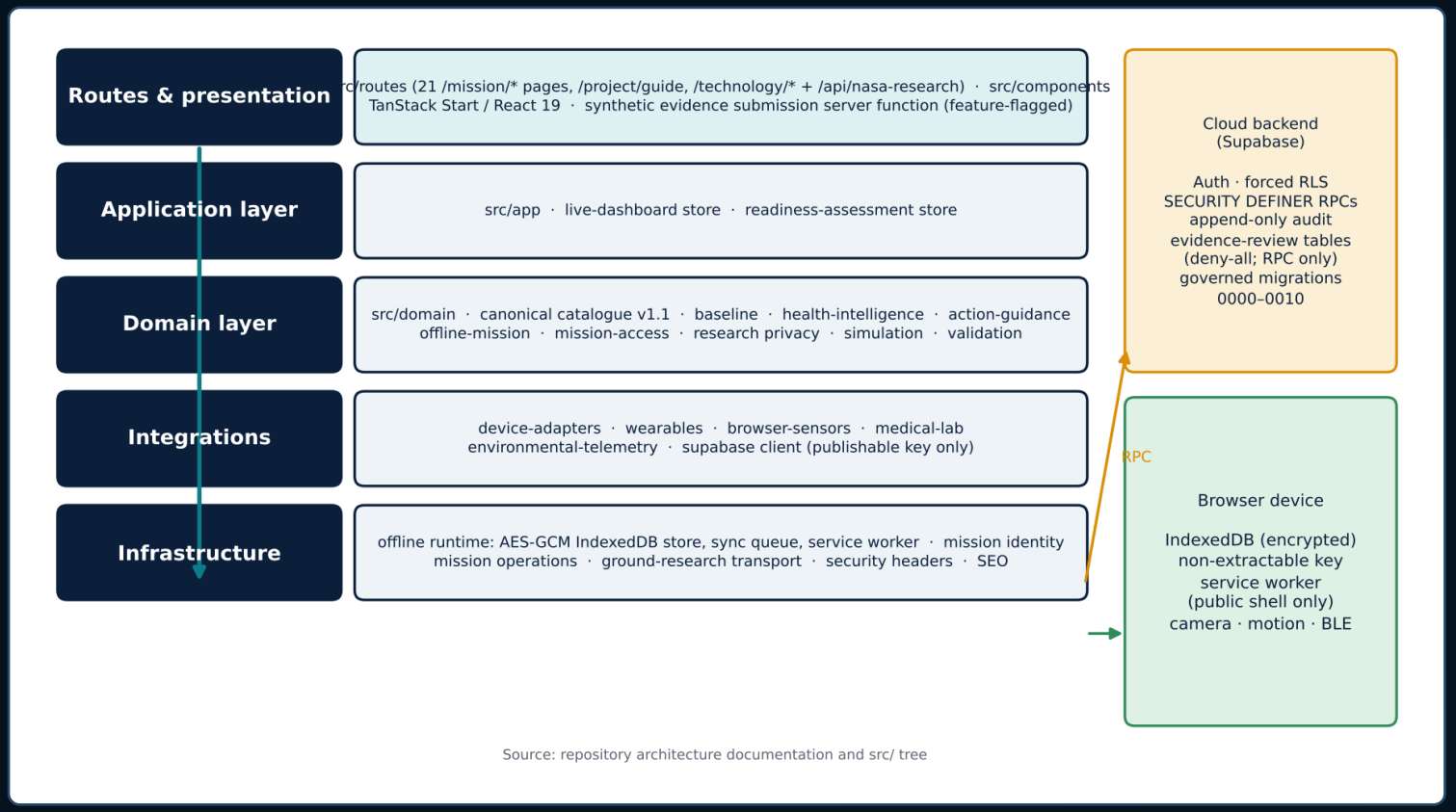
Status ledgers

- Validation and usability criteria
- Real-device acceptance cases
- Independent science gates
- UI worksheets never change them

Source: src/domain/scientific-localization; drizzle/migrations/0006–0010; data/validation/; data/submission/**

SYSTEM ARCHITECTURE

Architecture: the cloud is not in the sensor-to-astronaut path



Key facts

- 21 mission pages + NASA research API
- Server function: user token, flag-controlled
- Forced RLS; RPC-only writes
- Evidence tables deny-all
- Governed migrations 0000–0010

Source: docs/ARCHITECTURE.md; src/routes; src/lib; drizzle/migrations

Four mission-scoped roles — enforced in the database

	Crew Self	Mission Operator	Ground Medical	Researcher
Crew Self	one crew slot	allowed	forbidden	forbidden
Mission Operator	allowed	—	forbidden	allowed
Ground Medical	forbidden	forbidden	—	allowed
Researcher	forbidden	allowed	allowed	—

Same mission, both memberships active. Different missions: no restriction.
Enforced in PostgreSQL (mission_roles_compatible + trg_membership_role_compatibility).

Rules

- Different missions: independent
- GM invite: bind recipient + independent approval
- Reviewer ≠ submitter (user level)
- Revocation = immediate deactivation
- No super-administrator account by design

Source: supabase/migrations (multi-role hardening); src/domain/mission-access

Identity: clear sign-in, safe errors, self-service password change

Sign-in

- Field validation
- Generic inline error under password
- Password cleared after failure

Password change

- Online only
- Current password verified first
- ≥ 12 chars; must differ; must match

Invitations

- One-time, hashed, expiring
- Inline success / failure
- Unknown response = failure

Source: src/components/mission-identity.tsx; src/infrastructure/mission-identity.ts; tests/identity

Five menus, one search — the whole platform in reach

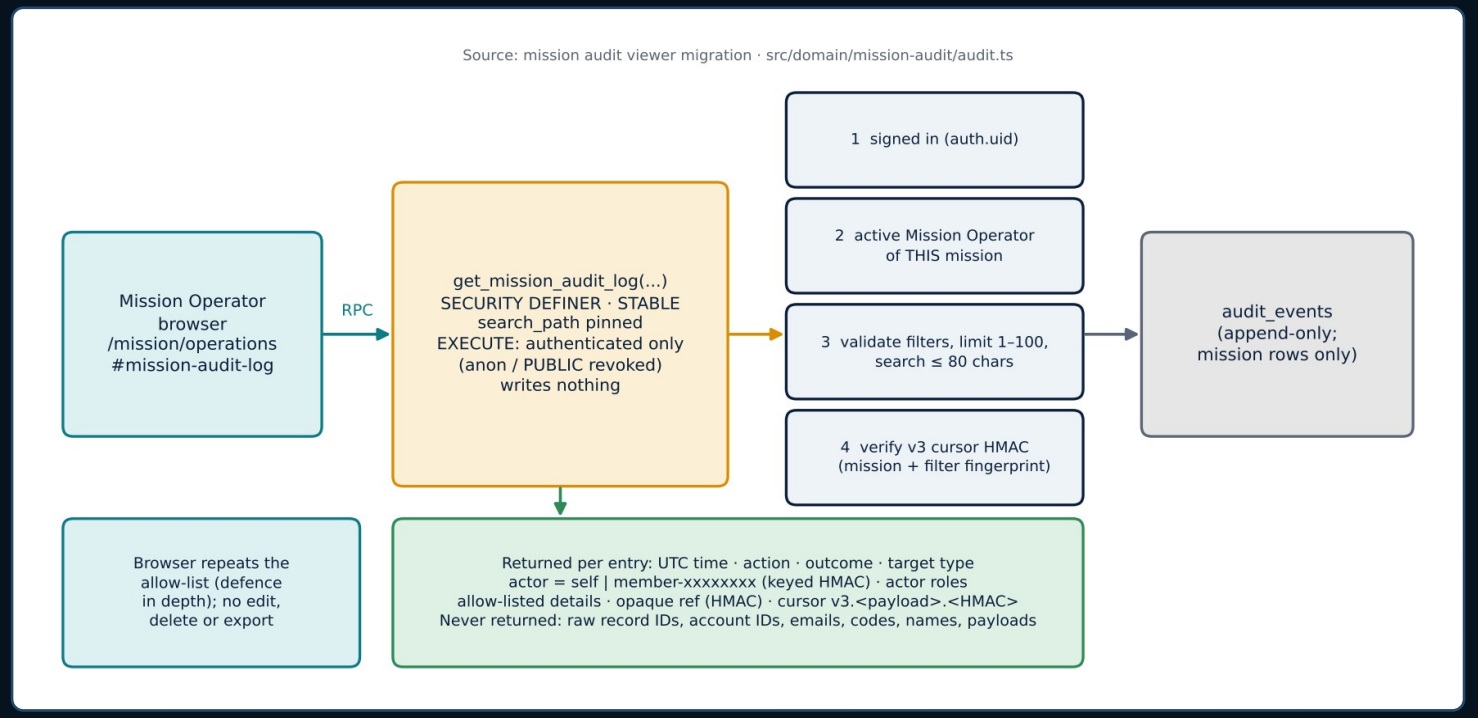


Navigation facts

- Home + 5 groups
- Search: Ctrl/⌘ + K
- Derived from one config
- Public labels only
- Stable URLs

Source: src/components/app-shell.tsx; navigation-search.tsx

Operators administer missions — and every action is audited

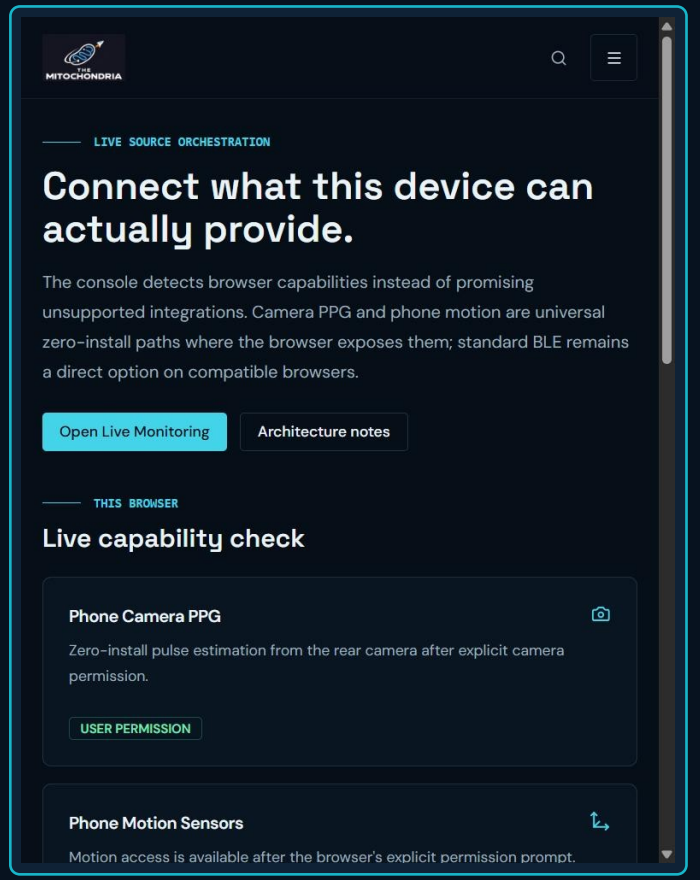


Feature	Status
Mission Settings (view)	FIELD-VERIFIED
Mission Settings (save)	OPERATIONAL
Audit Log	FIELD-VERIFIED
Crew, invites, memberships	FIELD-VERIFIED

Source: src/components/mission-operations.tsx, mission-settings-panel.tsx, mission-audit-log.tsx

Real sensors, honest states: nothing is ever fabricated

Source	Status	Notes
Camera PPG	FIELD-VERIFIED · one phone	Pulse estimate · not ECG
Phone motion	FIELD-VERIFIED · one phone	Clear error if no events
Web Bluetooth HR	OPERATIONAL · HW test pending	Standard Heart Rate Service
Health Connect / HealthKit	FUTURE · contract defined	Needs native host
Habitat & dosimetry	FUTURE · contract tested	No live feed yet



Device Integration Center — live site

Source: src/integrations/browser-sensors; real-device evidence record (data/submission)

Live monitoring separates identity, transport, sample and storage

Mission identity

AUTHENTICATED MEMBERSHIP · LOCAL FALLBACK

Device transport

Connected ≠ sample received

Freshness

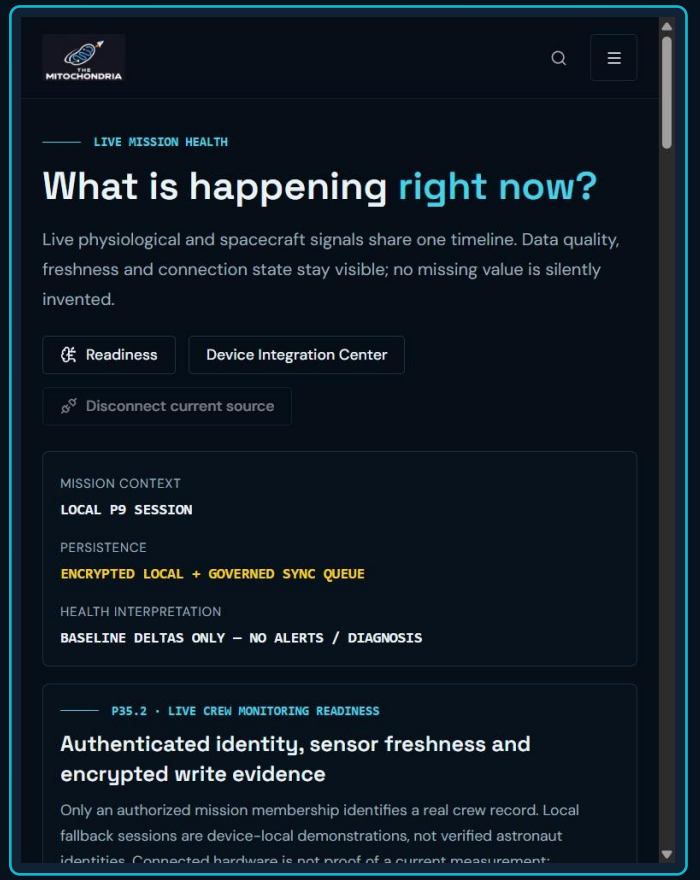
- LIVE · STALE · WAITING · DISCONNECTED
- 90 s physio / 180 s env
- Technical only

Last physical sample

Timestamp or NO SAMPLE

Encrypted persistence

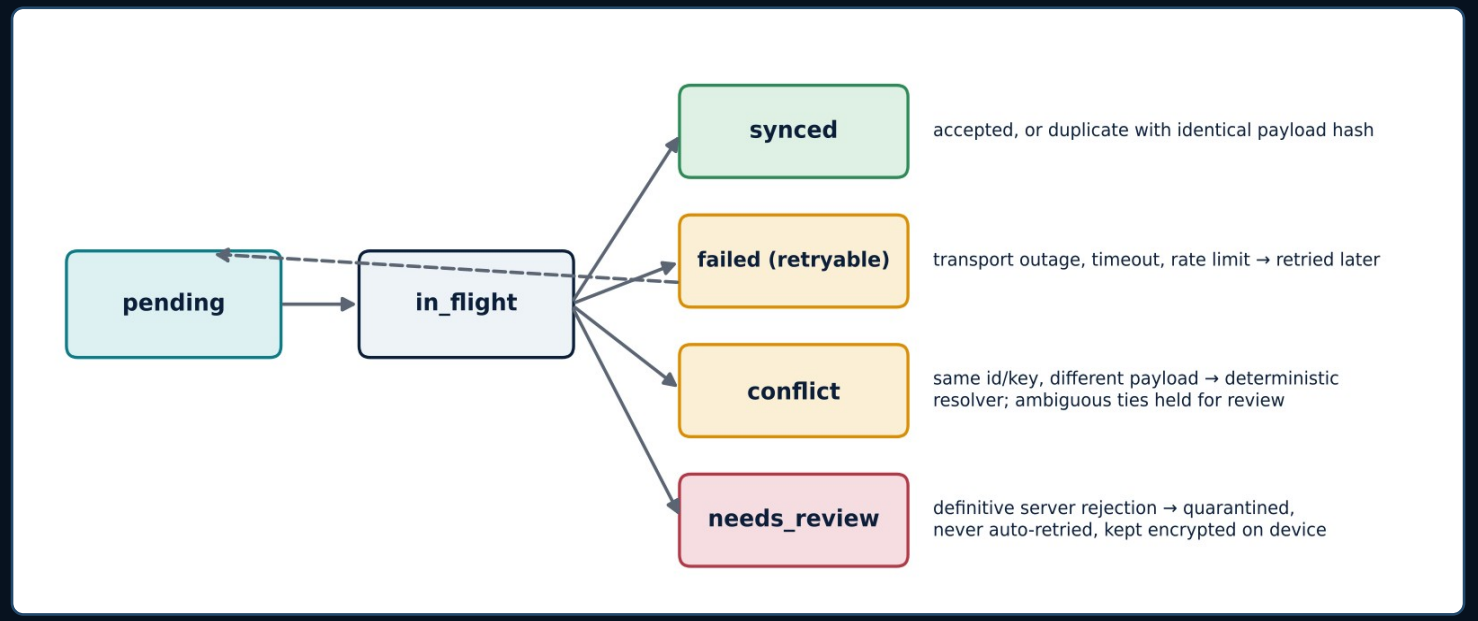
Confirmed count · WRITE FAILED



Live Monitoring — live site

Source: src/components/live-dashboard.tsx; src/app/live-dashboard/store.ts

Offline first: encrypt locally, synchronise with authority



Next verification: airplane-mode test on a physical device

Guarantees

- AES-GCM, non-extractable key
- Idempotency + payload hash
- No anonymous cloud writes
- Service worker: public shell only

Source: src/infrastructure/offline/; docs/OFFLINE_MISSION_MODE.md*

LABORATORY EVIDENCE INTAKE

Mission Documents: synthetic lab reports reviewed in the browser

1 Crew Self + consent

2 Local extraction

3 Exact name + unit

4 Human selection

5 Date + confirmation

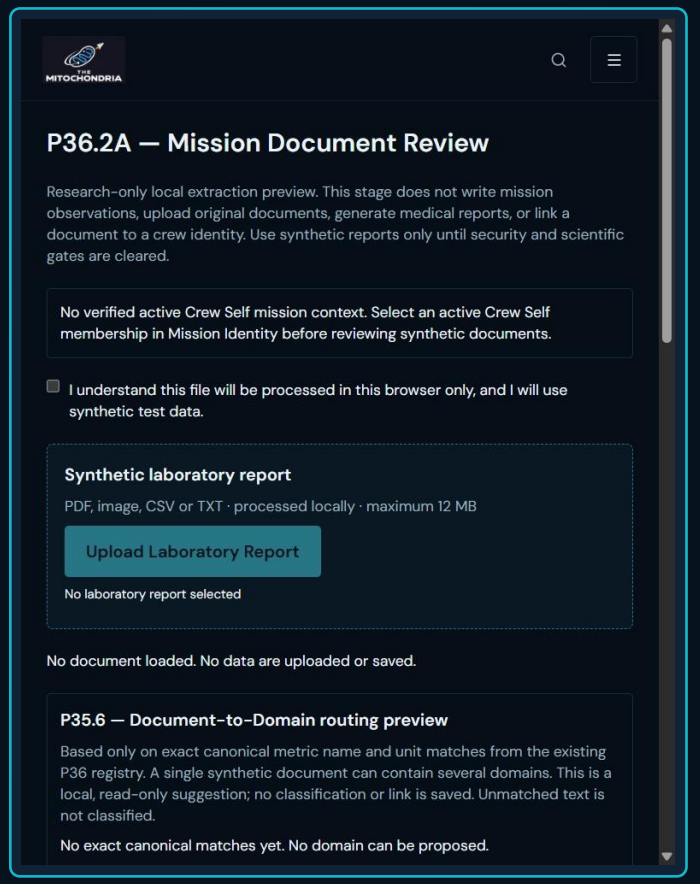
6 Preview or submit

Accepted input

PDF, image, CSV, TXT ≤ 12 MB · synthetic only

Never

Upload of the original · unit guessing

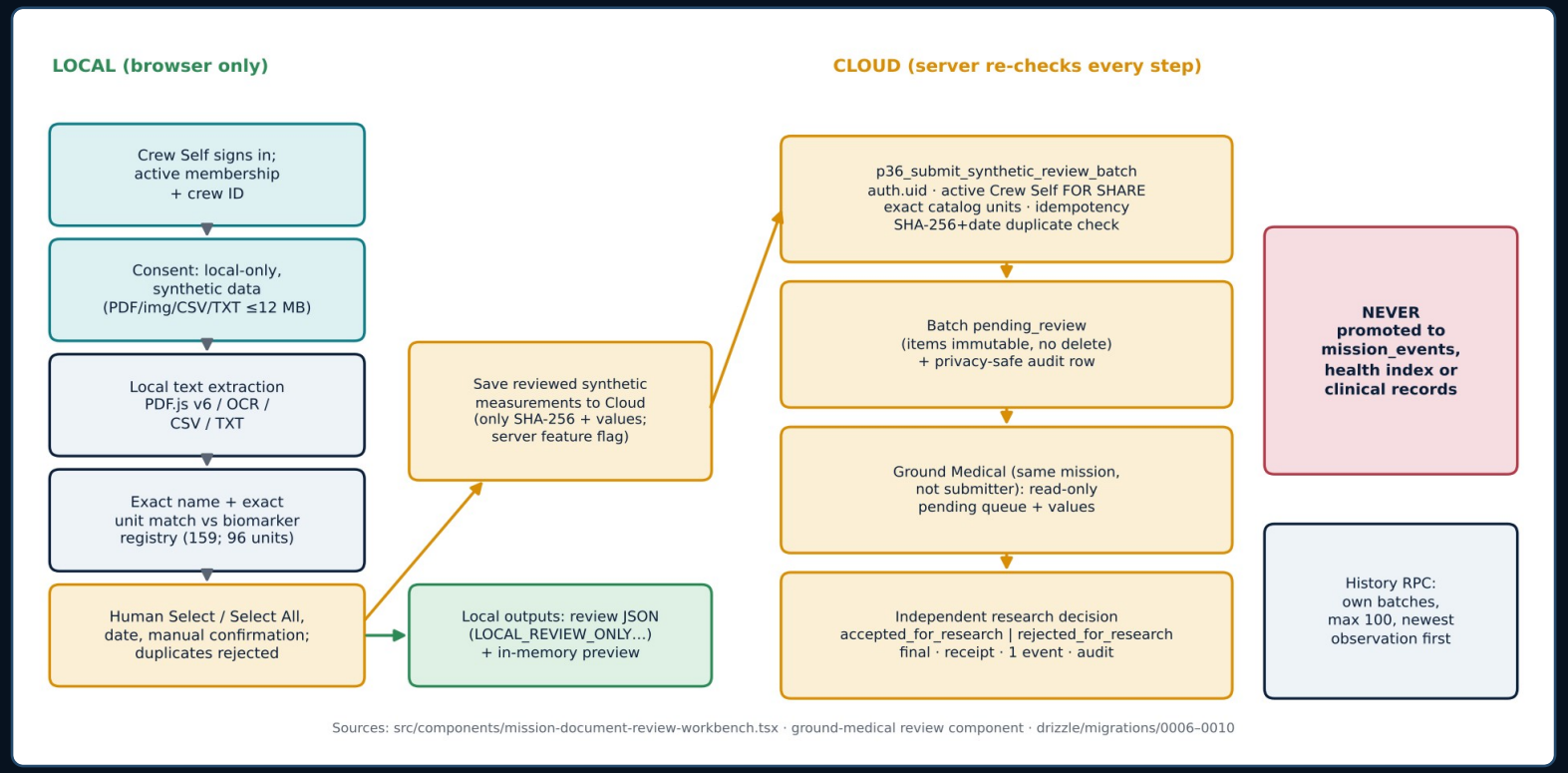


Mission Documents — live site

Source: src/components/mission-document-review-workbench.tsx; src/domain/biomarkers

CONTROLLED SYNTHETIC SUBMISSION

Synthetic Cloud batches: server-checked, idempotent, pending review



Server checks

- auth.uid + Crew Self lock
- 96 metric/unit pairs
- SHA-256 + date dedupe
- Idempotent replay
- pending_review only

Source: src/lib (submission server function); drizzle/migrations/0006–0008; tests/integration-pg

INDEPENDENT REVIEW

Independent Ground Medical review — research-only and final

Check	State
Read-only review functions	DEPLOYED
Decision function	DEPLOYED
Database integration suite	RECORDED 305 / 305
Independent live acceptance	PENDING verification

Decision guarantees

- Same mission, different user
- Locks; one event per batch
- Receipt; replay-safe
- Audit without values or notes
- Never promoted to health records

Source: Ground medical review component; drizzle/migrations/0009–0010; tests/integration-pg, tests/security

EXPLAINABLE HEALTH INTELLIGENCE

Health intelligence explains — it never diagnoses

Pattern	Evidence	Status
Sleep / fatigue / cognition	supported	non-diagnostic
Workload / recovery / autonomic	supported	non-diagnostic
Hydration / orthostatic	supported	non-diagnostic
CO_2 with symptoms	limited	contextual only
Bone unloading / renal	supported	non-diagnostic
Microbiome / immune	research only	never operational

Outputs

- INFORMATION · REVIEW
- DATA QUALITY · RESEARCH ONLY
- Operational alerts: DISABLED
- Self-assessment: session-only

Source: src/domain/health-intelligence/patterns.ts; src/components/explainable-alerts.tsx

NASA RESEARCH DATASET IMPORT

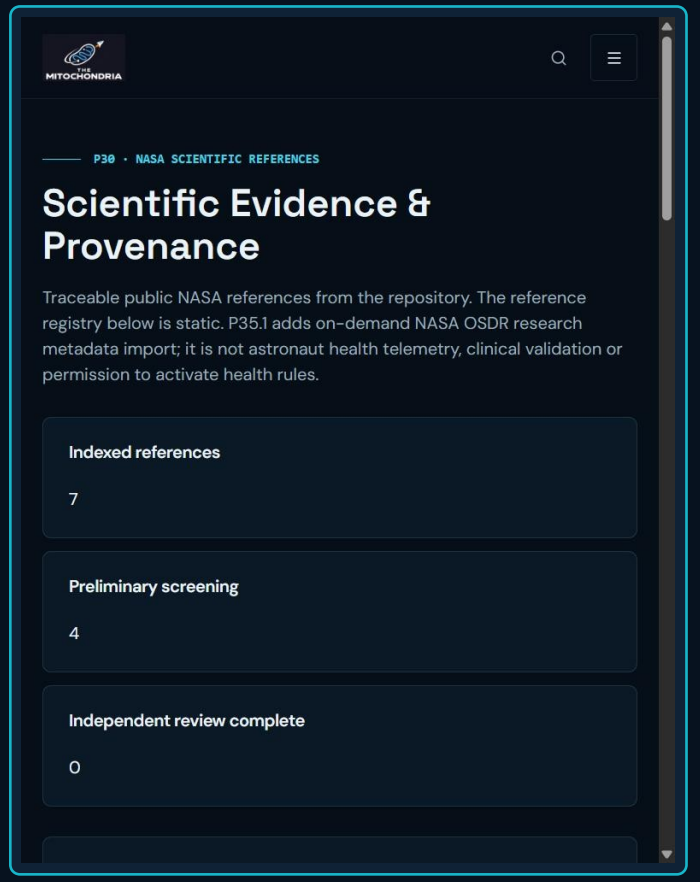
NASA OSDR: real research metadata, clearly not crew data

Bounded proxy

- GET /api/nasa-research
- Fixed origin; 12 s; 1 MiB
- Accession OSD-<n>; ≤ 500 rows
- No credentials; no-store

Labels on every package

- classification: research_only
- subjectBinding: none
- clinicalActivation: false
- Not astronaut telemetry



NASA Scientific Evidence — live site

Source: src/routes/api.nasa-research.ts; src/domain/nasa-research/import.ts

SYNTHETIC DEMONSTRATION ENVIRONMENT

A synthetic demo environment that cannot be mistaken for evidence

10-step journey

Advisory access hints

Tracker + runbook

Session-only; JSON

18 fictional observations

4 domains × 3 timepoints

Offline report

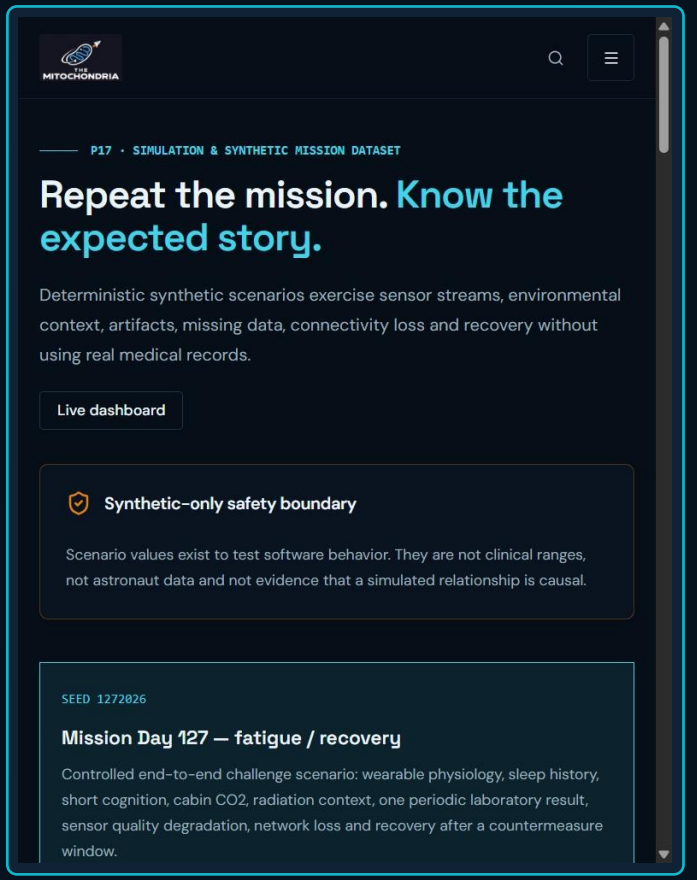
JSON · CSV · Print-PDF

Integrated bundle

Rejects tampering

Deterministic scenarios

Same seed → same fingerprint



Synthetic Mission Simulation — live site

Source: src/components/synthetic-mission.tsx; mission-demo-launchpad.tsx and sub-panels

EVALUATOR DEMONSTRATION FLOW

Evaluator demo in 15–20 minutes — no admin access, no real data

Path	Time	Steps
A · Public	~7 min	Home → Devices → Live → Scenario → NASA OSDR → Validation
B · Crew Self	~8 min	Sign-in → Guidance → Documents → Domains → Alerts → Offline
B · Operator	~2 min	Launchpad: tracker, preview, report, bundle
B · Ground Medical	~1.5 min	Review queue → View values

Evaluator Quick Start & Demonstration Guide — Final Competition Edition

Source: Evaluator Quick Start & Demonstration Guide, Section 5

Security and privacy are enforced where data lives

Controls

- Forced RLS; RPC-only writes
- Evidence tables deny-all
- Audit without values
- CSP, HSTS, Permissions-Policy

Supply chain

- Trivy, pinned actions
- HIGH findings remediated
- Patched pdfjs-dist 6.4
- Frozen lockfile builds

Still ahead

- Final security sign-off
- Provider-level finding
- Penetration test
- MFA

Source: docs/SECURITY_MODEL.md; src/security/headers.ts; .github/workflows/security.yml

Validation evidence — and what it does not prove

Evidence	Result
Full application gate	RECORDED 382 pass / 0 fail
Disposable PostgreSQL 17.9	RECORDED 305 / 305
Critical validation criteria	PASS 10 · battery = limitation
Independent build reproduction	PASS
Real-device capture	FIELD-VERIFIED · one phone
Human device cases (15 + 9)	PENDING verification
Independent science gates (8)	PENDING verification
Live review acceptance	PENDING verification



Not proven by tests

- Clinical validity
- Flight suitability
- Real-device generality
- Operational certification

*Source: Repository validation records; data/validation/**

Maturity at a glance — operational today, honest about the rest

Operational	Mission identity and roles, mission operations, audit log, live capture, encrypted offline store and governed sync, navigation and search	**AVAILABLE**
Real-device evidence	Camera PPG pulse estimate and phone motion recorded on a physical phone; Bluetooth heart-rate path implemented but not yet hardware-tested	**ONE DEVICE**
Automated test evidence	382 application checks passing; 305/305 database assertions on disposable PostgreSQL 17.9 (repository validation records, Oct 2026)	**RECORDED**
Synthetic demonstration	Mission Day 127 and multi-crew scenarios, 18-observation demo dataset, offline report, synthetic document batches	**SYNTHETIC**
Research-only science	Handbooks (bone 63, cardiovascular 96), immune and behavioural draft catalogues (35 + 36), NASA OSDR metadata — reference only	**REFERENCE**
Pending verification	Live acceptance of the ground medical review workflow; human real-device acceptance cases; independent scientific review; platform security gate	**PENDING**
Future capability	Persisted personal baselines; native Health Connect / HealthKit hosts; habitat telemetry feed; governed administrator role	**PLANNED**
Clinical validation	Not claimed. Clinical activation is disabled; outputs are decision support for qualified humans only	**NOT CLAIMED**

Boundaries

- Research prototype
- Synthetic data only
- Not clinically validated
- Not flight-qualified
- Not NASA-endorsed

Source: Master Manual §1.2, §32 and Appendix A

Next: independent verification, then a governed future

NOW

Live acceptance of the review workflow

NEXT

Human real-device acceptance

THEN

Independent science gates

ON ARRIVAL

Scientific review documents → reconcile 71 proposals

PRODUCT

Persisted baselines · governed admin · native hosts · habitat feed

FUTURE SCIENTIFIC UPDATE — reserved: no content from the pending review documents is anticipated or incorporated.

*Source: Master Manual §39; data/validation/**

CLOSING

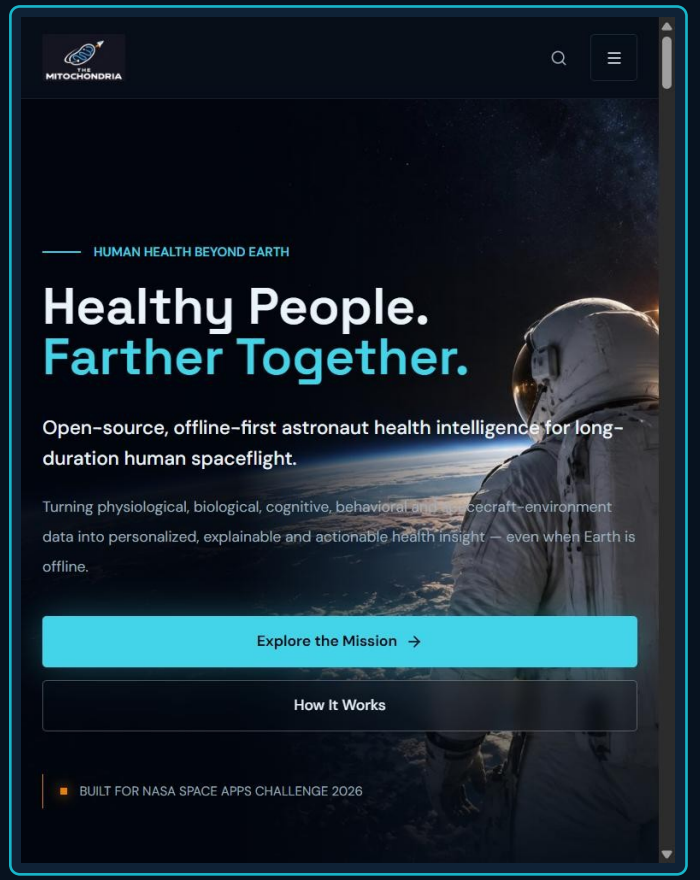
Healthy people. Farther together — with evidence you can check.

Try it

- mitochondria.edocinfo.com
- Public path: no account
- Evaluator guide: 15–20 min

Read and review

- Master Operations & Technical Manual
- github.com/computech-gr/astro-vitality-core
- Apache-2.0 · contributions welcome



Home — live site

Research prototype · synthetic data only · not NASA-endorsed · not clinically validated